
ECON 312 — Intermediate Microeconomics

Handout: Budget Constraints

Fall 2026

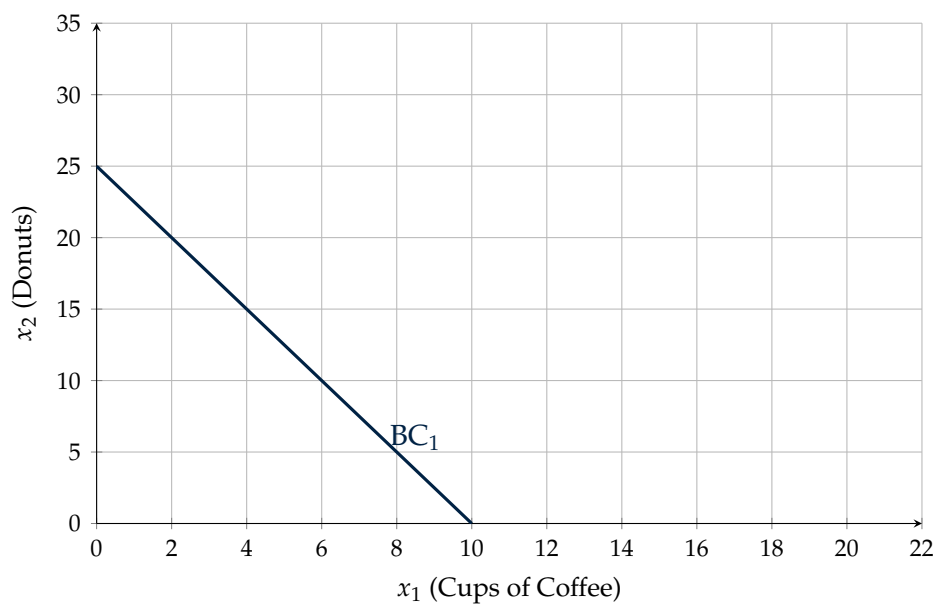
Baseline Budget Set: A consumer allocates a fixed budget of $m = \$50$ between two goods: Coffee (x_1) priced at $p_1 = \$5$ per cup, and Donuts (x_2) priced at $p_2 = \$2$ per donut.

Baseline Budget Equation: $5x_1 + 2x_2 = 50$

Part 1. Changing Income (Shifts)

On the grid below, the baseline budget line is already provided. Graph the two following changes on this same coordinate system:

1. **Budget Expansion:** An increase in nominal income to $m' = \$60$ (prices constant).
2. **Budget Reduction:** A decrease in nominal income to $m'' = \$30$ (prices constant).

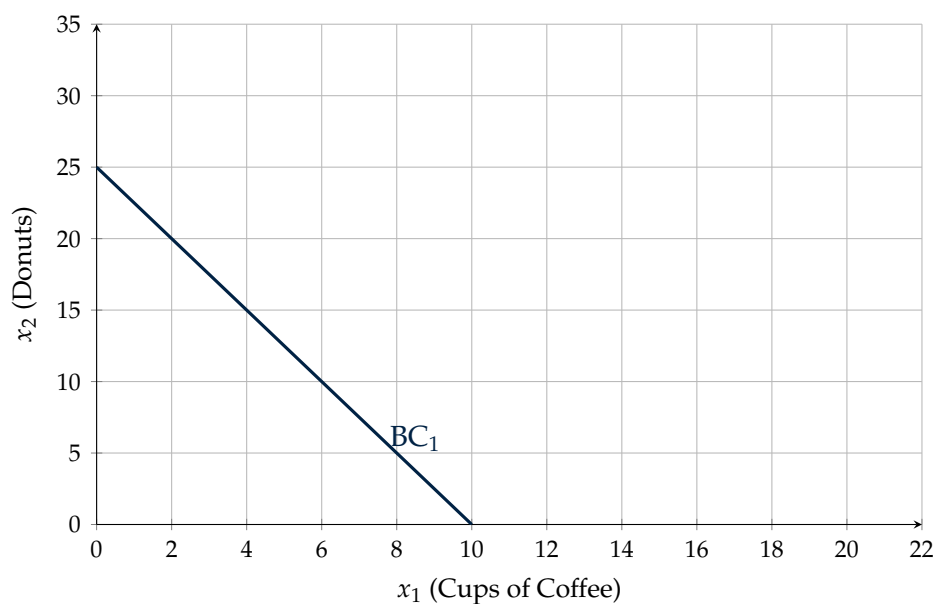


	Income Increase (\$60)	Income Decrease (\$30)
New Horizontal Intercept	_____	_____
New Vertical Intercept	_____	_____
New Slope	_____	_____
Direction of Shift	_____	_____

Part 2. Changing Relative Prices (Pivots)

Return to the baseline scenario ($m = \$50$, $p_2 = \$2$). Graph the two following price changes on this same coordinate system:

1. **Price Increase:** The price of coffee increases to $p'_1 = \$10$ per cup.
2. **Price Decrease:** The price of coffee decreases to $p''_1 = \$2.50$ per cup.



	Price Increase (\$10)	Price Decrease (\$2.50)
New Horizontal Intercept	_____	_____
New Vertical Intercept	_____	_____
New Slope	_____	_____
Effect on Opportunity Cost of x_1	_____	_____

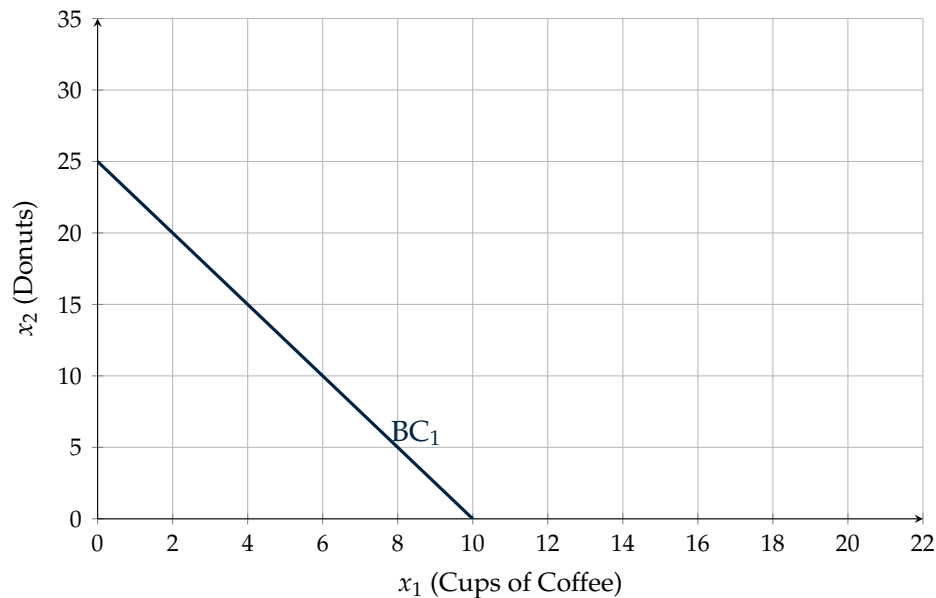
Shifts vs. pivots

An increase in nominal income m expands the budget set via a _____ shift. An isolated increase in p_1 shrinks the budget set by _____ the line inward around the x_2 -intercept. A change in p_1 alone always pivots around the _____-intercept, never the x_1 -intercept.

Part 3. Government Intervention I: Quantity Tax vs. Ad Valorem Tax

Return to the baseline scenario ($m = \$50$, $p_1 = \$5$, $p_2 = \$2$). The government now taxes coffee. Graph both policies on the same coordinate system as the baseline:

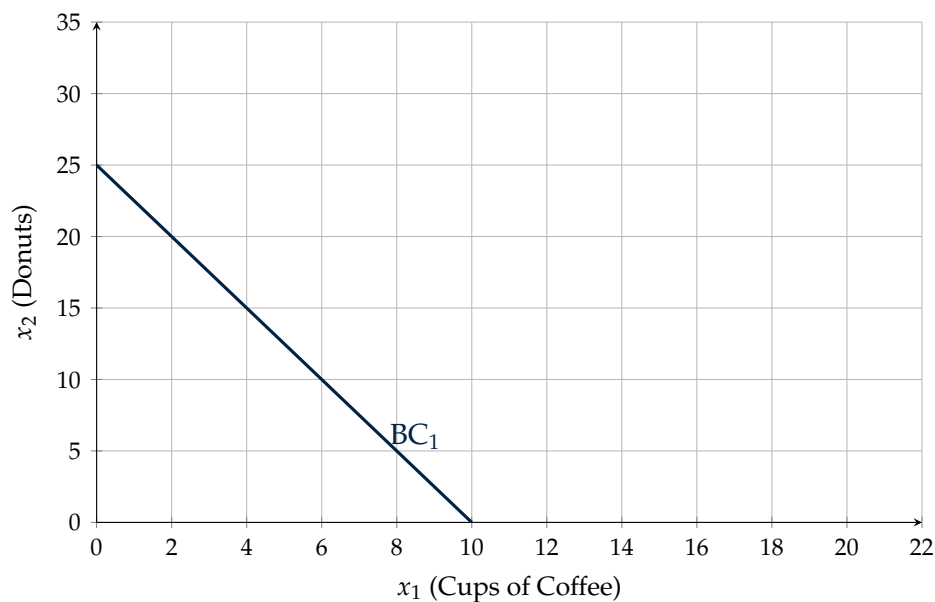
- 1. **Quantity (Specific) Tax:** A per-unit tax of $t = \$3$ is levied on each cup of coffee, so the consumer pays $p'_1 = p_1 + t$.
- 2. **Ad Valorem Tax:** A $\tau = 20\%$ tax is levied on the *value* of each coffee purchase, so the consumer pays $p''_1 = p_1(1 + \tau)$.



	Quantity Tax ($t = \$3$)	Ad Valorem Tax ($\tau = 20\%$)
Effective Price of Coffee, p'_1	_____	_____
New Horizontal Intercept	_____	_____
New Vertical Intercept	_____	_____
New Slope	_____	_____

Part 4. Government Intervention II: Quantity Subsidy

Return to the baseline scenario. The government now *subsidizes* coffee at $s = \$1$ per cup, so the consumer effectively pays $p'_1 =$ _____.



Quantity Subsidy ($s = \$1$)

Effective Price of Coffee, p'_1 _____

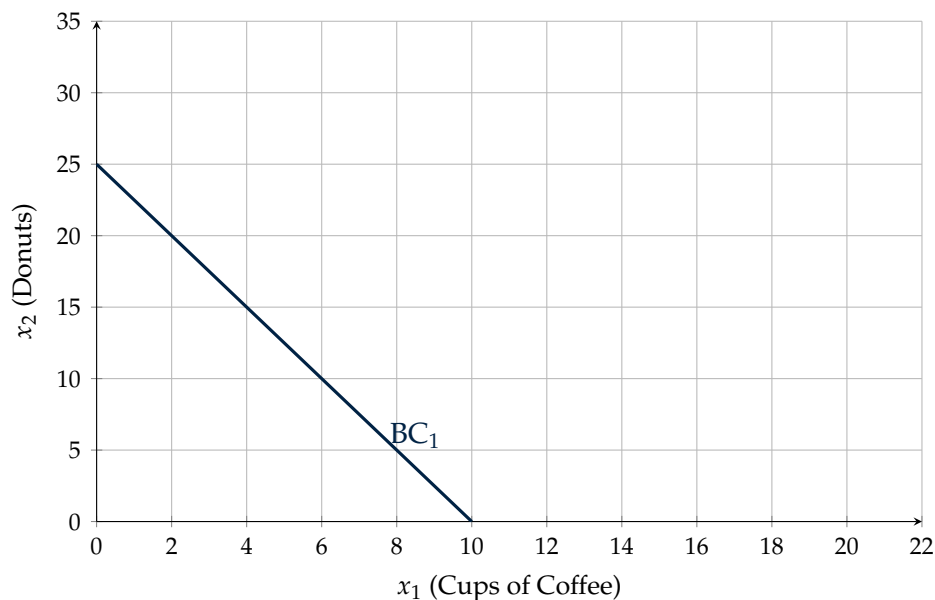
New Horizontal Intercept _____

New Vertical Intercept _____

Direction of Pivot _____

Part 5. Government Intervention III: Lump-Sum Tax

Return to the baseline scenario. The government instead levies a **lump-sum tax** of $T = \$10$, collected regardless of the consumer's purchases, so disposable income falls to $m' =$ _____.



Lump-Sum Tax ($T = \$10$)

New Income, m' _____

New Horizontal Intercept _____

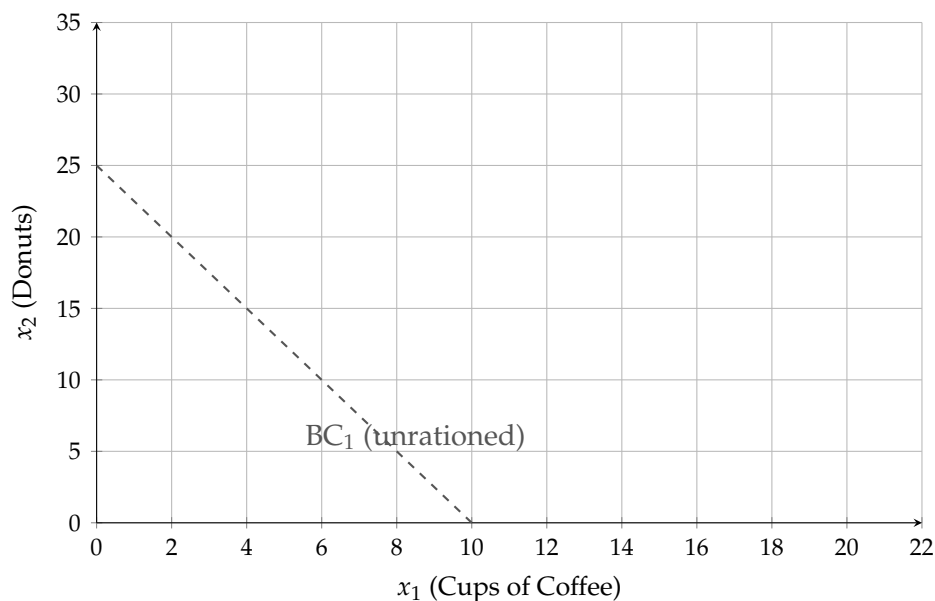
New Vertical Intercept _____

New Slope _____

Part 6. Non-Price Rationing

Return to the baseline scenario ($5x_1 + 2x_2 = 50$). Suppose the government does *not* touch prices or income at all. Instead, it imposes a physical ration: no consumer may purchase more than $\bar{x}_1 = 6$ cups of coffee, full stop.

On the grid below, shade (or clearly outline) the **feasible consumption set** under rationing.

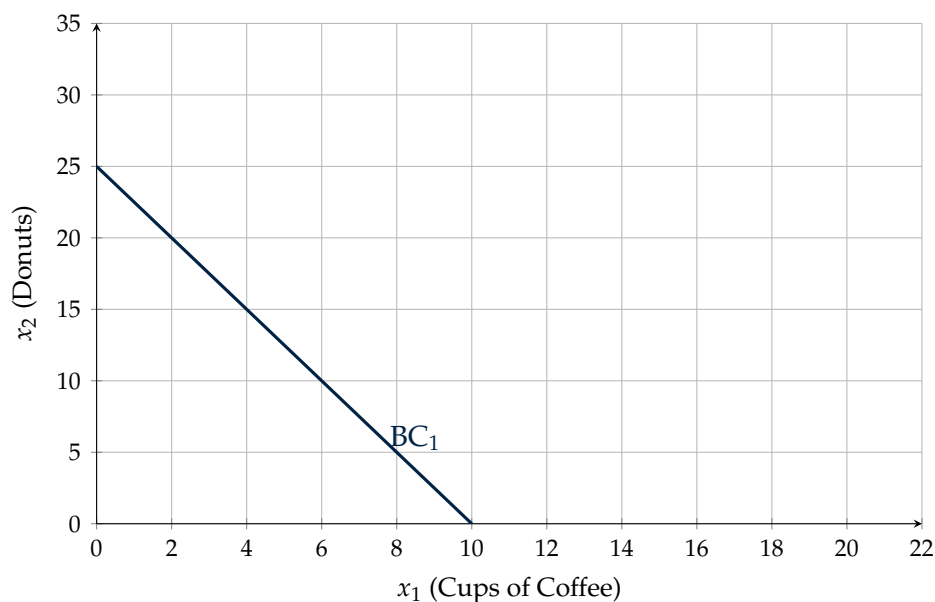


Coordinates of the kink point $(\bar{x}_1, x_2)$ _____

Is the point $(8, 5)$ feasible? Why or why not? _____

Part 7. Government Intervention V: In-Kind Transfer (SNAP-Style Benefit)

Return to the baseline scenario ($5x_1 + 2x_2 = 50$). Suppose the government now provides an **in-kind transfer** of $G = 10$ donuts, at no cost to the consumer. These donuts cannot be resold or traded for coffee. Graph the new budget frontier on the same coordinate system as the baseline, and compare it to a hypothetical **cash-equivalent** transfer of \$20 (i.e. $G \cdot p_2$).



	In-Kind Transfer ($G = 10$)	Cash Equivalent (\$20)
New Vertical Intercept	_____	_____
New Horizontal Intercept	_____	_____
New Slope	_____	_____
Is the in-kind BC a subset of the cash BC?	_____	